

Dr. Ashish K Sharma

Professor of Physics

Department of Physics & Astronomical Sciences, Central University of Jammu, J&K – 181143, India
ashish.phy@cuammu.ac.in · <https://dr-aks.github.io> · ORCID 0000-0002-5240-3472

Ashish K Sharma is Professor of Physics at the Central University of Jammu and heads its Energy Materials and Devices Laboratory. A doctorate at IIT Delhi in 2013 was followed by eight years at the Inter-University Accelerator Centre, as DST-INSPIRE Faculty Fellow and then SERB Research Scientist, and by a period at UPES Dehradun, before Jammu in 2023.

Energy comes first: thermoelectric materials in inorganic, organic, composite, flexible and printable form, built into devices that draw electricity from waste heat. The line has won four funded projects, the RSC Advances Outstanding Student Paper Award in 2024 and a Best Ph.D. Thesis Award in 2025.

Second, devices on strategic materials: gallium nitride, gallium oxide and silicon carbide, for high power, high temperature, low noise and radiation-harsh service, improved by passivation, annealing and processing, and tested for radiation tolerance. Both thrusts rest on defect formation and evolution, studied by experiment and theory together, now carried into one- and two-dimensional nanomaterials and heterostructures for space.

Much of the apparatus is his own: low-cost Seebeck, resistivity, thermal conductivity, gas-sensing and in-situ DLTS systems, several with few equivalents in the country.

The problems follow the national missions: devices for the Semiconductor Mission, low-noise measurement and defect spectroscopy for the Quantum Mission, radiation-hard electronics and deep-ultraviolet detectors for the space programme, and instruments built here for Make in India.

He has 91 papers, three patents, an h-index of 27 and close to 1,860 citations, on ₹2.05 crore from DST, SERB, ISRO, DRDO and IUAC, and in 2019 was among fewer than 600 young scientists chosen worldwide for the Lindau Nobel Laureate Meeting. Seven doctoral scholars and over twenty master's students have worked with him, and value his teaching and his readiness to help as much as the physics.

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